Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Average velocity

Meaning in my words How would you explain average velocity to a friend?	**How can you tell?** A cart travels out and back. Which two points determine its average velocity?
Example or sketch Right is positive. A cart goes from +8 m to +2 m in 3 s. What is its average velocity?	**Non-example and why** “Average the two speeds, and you always get average velocity.” What is missing?

Average speed

Meaning in my words How would you explain average speed to a friend?	**How can you tell?** On a trip out and back, which distance goes on top of the fraction?
Example or sketch You walk 3 m out and 3 m back in 4 s. What is your average speed?	**Non-example and why** “I got back to the start, so my average speed was zero.” What would you say?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Slope

Meaning in my words How would you explain slope to a friend?	**How can you tell?** On a position–time graph, what does a line falling to the right tell you?
Example or sketch A line goes from 2 m at 0 s to 8 m at 3 s. What slope would you label?	**Non-example and why** The graph reaches 8 m. Does that make the slope 8 m/s?

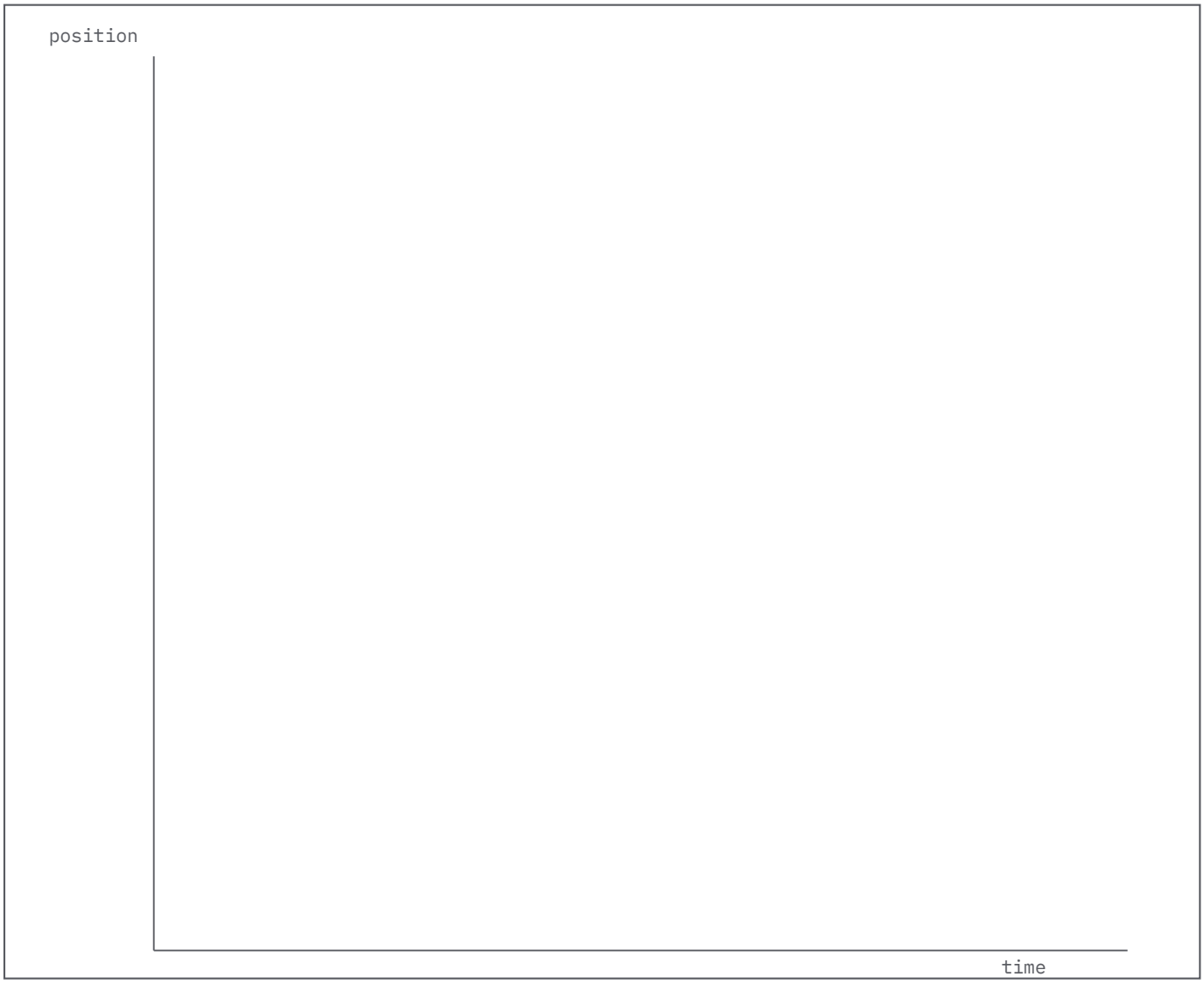
Instantaneous velocity

Meaning in my words How would you explain instantaneous velocity to a friend?	**How can you tell?** On a curved position–time graph, what line would you draw at exactly 2 s?
Example or sketch What does a tangent slope of −3 m/s at 2 s tell the camera operator?	**Non-example and why** A partner joins the first and last points of a curved trip. Which velocity does that slope give?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch a falling position–time line. Label the signed rise, positive run and velocity unit.



My marks mean

Is it really slowing down?

03

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use a position-time graph to find velocity.

CHOOSE ONE JOB TOGETHER

RADAR OPERATOR ☐

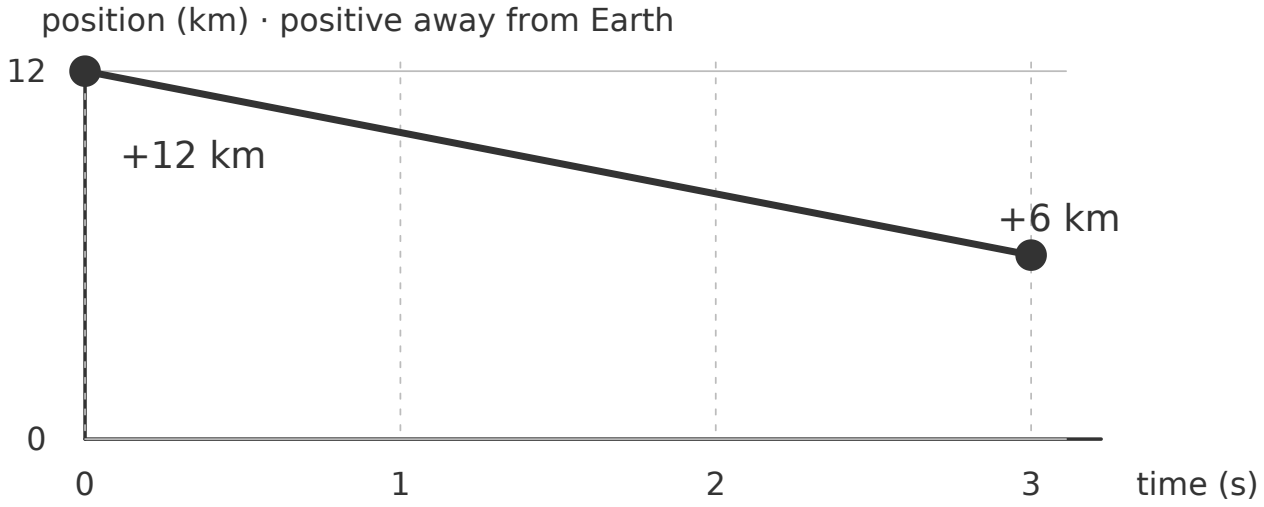
Calculate the signed average velocity. Mark the two sightings on the graph so the next operator can check your result.

TRAFFIC BROADCASTER ☐

Report a speed and direction. Tell viewers whether these two sightings show that XJ is slowing down.

Point to a clue. Keep your first marks; add the update using the color key.

Radar sees XJ at +**12 km**, then +**6 km** three seconds later. Positive points away from Earth.



01 OUR FIRST CALL

What can you report about XJ's motion? Use the graph to support your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Write the correction you would send or say on air.

Switch speakers. What can these two sightings tell you about average velocity? What can they not tell you about motion between sightings?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks